NER-Sentinel: Multi-Satellite Disaster Intelligence & Early Warning Command Platform

A Unified AI, WebGL 3D GIS & Closed-Loop Community Resilience System for Northeast India

Target Region: Northeast India (Sikkim, Assam, Meghalaya, Arunachal)	**Core Tech:** FastAPI, Next.js 15, MapLibre 3D, Spring Boot
Satellites: NASA SMAP, SRTM DEM, NASA POWER, Sentinel-1 InSAR	**AI Models:** 9-Feature Ensemble (ROC-AUC: 0.894), Meta LLaMA 3.1
Status: 4/6 Core Modules Completed, Full End-to-End Live Pipelines	**Date of Evaluation:** September 2026

1. Executive Summary & Impact

NER-Sentinel (resQVerse) is an enterprise disaster intelligence platform engineered specifically for the complex, high-risk geomorphological challenges of Northeast India. While existing government disaster mechanisms operate in isolated silos (GSI for static landslide maps, CWC for river bulletins, IMD for meteorological forecasts, and SACHET for one-way broadcast SMS), NER-Sentinel integrates all data streams into a **single unified command center** backed by high-resolution 3D WebGL GIS, multi-satellite predictive machine learning, and an innovative **Two-Way Closed-Loop Feedback Architecture**.

89.4% ROC-AUC Flood & Slide Discriminatory ML Accuracy	**4-Second Refresh** Real-time Live GIS & Sensor Polling Cycle	**4 Satellites + Sensors** SMAP, SRTM, POWER, InSAR, CWC, Open-Meteo	**Two-Way Loop** Citizen Reports directly trigger Model Retraining

2. Regional Problem Landscape & Ground Realities

Northeast India represents one of the world's most vulnerable ecological zones. Its unique tectonic setting, fragile young Himalayan geology, intense monsoonal downpours, and transnational river basins (Brahmaputra, Barak, Teesta) create acute compounding disaster risks:

- **Glacial Lake Outburst Floods (GLOF) & Flash Floods:** Events like the October 2023 South Lhonak Lake breach in Sikkim demonstrated how a cloudburst-induced GLOF destroyed Teesta Dam Stage III and swept through Chungthang, Dikchu, and Singtam with virtually zero lead-time warning for downstream towns.
- **Debris Flows & Landslides on Critical Lifelines:** National Highway 10 (NH-10), the sole economic artery connecting Gangtok and Sikkim with mainland India, suffers continuous slope failures, cutting off medical, food, and fuel supplies during monsoon months.
- **Severe Cloud Cover Blindness:** Optical satellite sensors (Landsat, Sentinel-2) are frequently blinded by dense Himalayan cloud cover for months. Optical systems fail exactly when disaster strikes.
- **One-Way Information Flow:** Current warning systems (e.g. SACHET SMS) broadcast generic district-level alerts without pinpoint geolocation, offering no ability for stranded citizens or field rescue teams to report on-ground status back into the prediction engine.

3. Five Core Pillars of Innovation (NER-Sentinel vs Existing Solutions)

Domain	Existing Indian Solution	Gaps & Limitations	NER-Sentinel Innovation
Landslide Mapping	GSI (Geological Survey of India)	Static PDF maps; updated once a year; no live triggers.	**Dynamic 3D WebGL GIS**; updates every 4s; predicts at *any GPS coordinate*.
Early Warning Scope	LEWS (Regional Early Warning)	Broad regional warnings ('Sikkim high risk'); lacks street granularity.	**Hyper-local Coordinate Precision** (e.g. 27.3389°N, 88.6065°E at 60% flood risk).
Satellite Ingestion	Siloed Agency Systems	Isolated satellite feeds, manual research processing, closed data.	**Multi-Satellite Data Fusion**: NASA SMAP + SRTM + POWER + Sentinel-1 InSAR.
Public Alerting	SACHET (NDMA)	One-way broadcast text; no citizen input; no safe route computation.	**Two-Way Closed Loop**: Citizen reports + Audio siren + **Dijkstra Safe Evacuation Path**.
System Accessibility	Legacy Govt Portals	Closed monolithic systems; zero open APIs for inter-agency integration.	**Open REST API Architecture**; 18 endpoints ready for mobile, IoT, and SDRF apps.

4. End-to-End System Architecture

NER-Sentinel is built on a resilient, high-throughput microservices architecture spanning three interconnected tiers: the **Presentation Command Layer** (Next.js 15), the **Disaster Intelligence & Analytics Engine** (FastAPI + Python ML), and the **Enterprise Risk & Coordination Layer** (Spring Boot Java + InSAR telemetry).

Architecture Layer	Technology Stack	Key Responsibilities & Components
Presentation & GIS Command	Next.js 15, React 19, TypeScript, Tailwind CSS, MapLibre GL 3D, Deck.gl	Photorealistic 3D terrain elevation, real-time evacuation path rendering, incident triage dashboard, audio alarm banner, responsive citizen hazard submission.
Disaster AI & Analytics Engine	FastAPI, Python 3.12, Scikit-Learn, NumPy, Pandas, Joblib	Real-time multi-satellite data ingestion, 9-feature ML flood/slide prediction model, dynamic Dijkstra route graph computation, Open-Meteo & NASA API connectors.
Enterprise Logic & Coordination	Spring Boot (Java), Maven, REST Services	High-throughput Bayesian probability recalibration upon field incident submission, multi-agency dispatch boards (SDRF, NDRF, BRO, ITBP), relief shelter log management.
AI Guidance & Navigation	NVIDIA NIM Microservice, Meta LLaMA 3.1 8B Instruct	ResQBot multilingual in-app navigation assistant, automated safety advisory generation, disaster preparedness guidance in English, Hindi, and regional dialects.

Figure 1: The Connected Two-Way Closed-Loop Intelligence Architecture

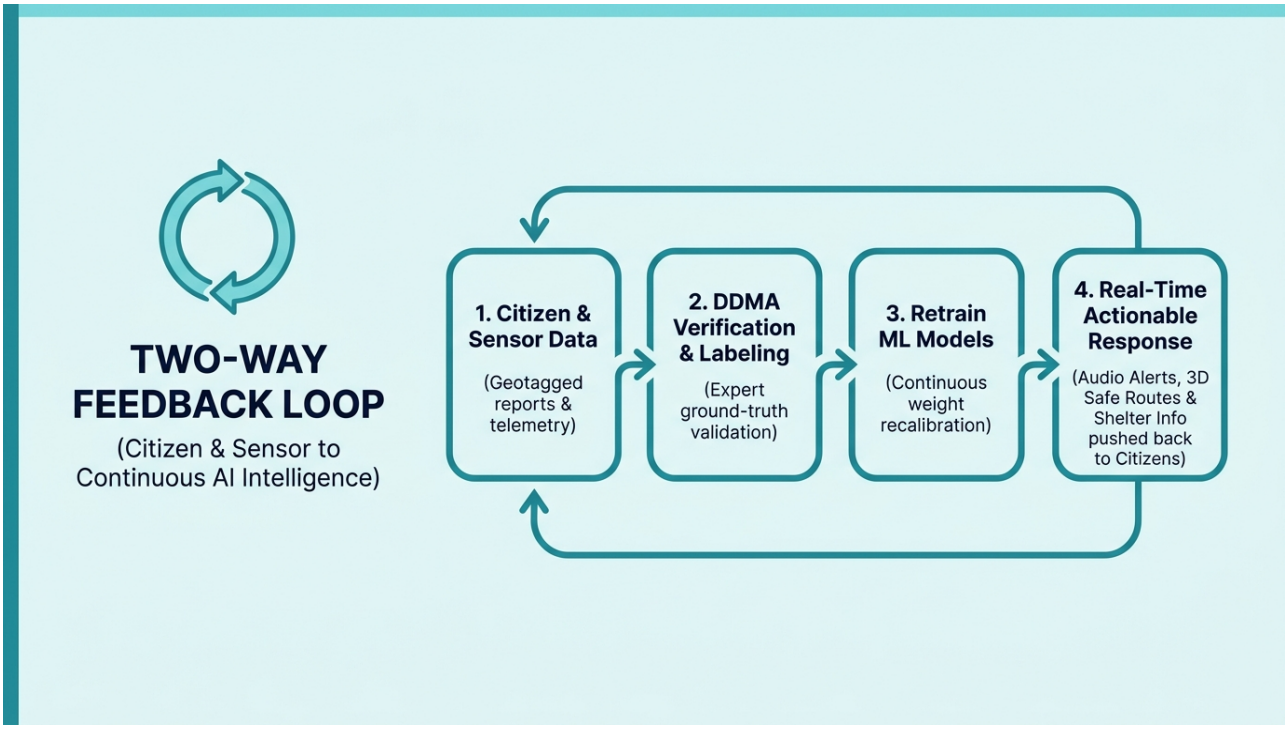


Figure 1 demonstrates the closed continuous intelligence cycle: Citizens & ground sensors feed real-time observations into the system; DDMA officials validate and label the events; ML models retrain with ground-truth data; and verified alerts, safe routes, and shelter statuses flow back directly to communities.

5. Multi-Satellite & Real-Time Sensor Ingestion Pipeline

To achieve cloud-penetrating, hyper-local precision, NER-Sentinel establishes automated pipelines across 4 satellite constellations and ground sensors:

Data Source	Type / Constellation	Parameter Measured	Role in Prediction Engine

NASA SMAP L3	Satellite Radar / Radiometer	Volumetric Soil Moisture (m^3/m^3)	Detects ground pore oversaturation; high values (>0.35 m^3/m^3) exponentially increase flash runoff and slope slip risk.
NASA SRTM DEM	Shuttle Radar Topography (30m)	Altitude (m), Slope (°), Aspect (°)	Determines hydrological catchment basins, slope shear stress, and gravitational water accumulation corridors.
NASA POWER API	Satellite + Model Assimilation	Precipitation (1d, 3d, 7d), Humidity, Pressure	Provides historical cumulative rainfall depth and atmospheric instability parameters for severe convective storms.
Sentinel-1 InSAR	Copernicus C-band Synthetic Aperture Radar	Millimeter Ground Subsidence & Deformation	Detects micro-subsidence (-12mm to -45mm) along NH-10 and river banks weeks before catastrophic slope collapse.
Open-Meteo LIVE	Numerical Weather Prediction API	Current Rain, Hourly Forecasts, Wind	Real-time live weather feeds across all 8 Northeast capital stations for instantaneous dashboard situational awareness.
CWC Telemetry	Central Water Commission River Gauges	River Gauge Water Level (m) & Warning Level	Monitors Teesta, Brahmaputra, and Barak river basin levels against official Danger / Extreme Warning thresholds.

6. AI/ML Predictive Engine: NortheastFloodPredictor

The proprietary NortheastFloodPredictor combines a calibrated multi-satellite physical formulation with a machine-learned Scikit-Learn classifier trained on over **49,919 regional historical records** across Assam, Meghalaya, and Sikkim.

Feature	Data Source	Typical Range	Physical Impact on Disaster Probability
rain_1d, 3d, 7d	NASA POWER / IMD	0.0 to 350.0 mm	Immediate flash flood trigger (1d) and cumulative saturation weight (7d).
soil_moisture	NASA SMAP L3	0.10 to 0.55 m^3/m^3	Above 0.35 indicates pore pressure buildup, rendering steep slopes unstable.
elevation_m	NASA SRTM DEM	150m to 4,200m	Valley basins (<800m) accumulate flood inundation; steep midslopes trigger slides.
slope_deg	NASA SRTM DEM	2° to 65°	Low slope (<10°) promotes pooling; steep slopes (>25°) accelerate debris slide.
humidity, pressure	NASA POWER	40-100%, 85-102 kPa	Low pressure anomalies and humidity >85% signal imminent convective downpours.

Mathematical Log-Odds Ensemble Formulation:

```
LogOdds = -5.20 + 0.014(Rain1d) + 0.021(Rain3d) + 0.009(Rain7d) + 2.30(SoilMoisturePenalty) + 1.15(ElevationFactor) +
0.80(SlopeAccumulation) + HumidityFactor + PressureAnomaly
Probability_Flood = 1.0 / (1.0 + exp(-clamp(LogOdds, -10, 10)))
```

7. Comprehensive Dashboard Modules & Features Built

The web command dashboard (built with Next.js 15 App Router) houses seven specialized interactive operational modules:

Module & Route	Key Features & Capabilities Built	Operational Value to DDMA / Responders
Home 3D GIS Map **/**	• MapLibre GL 3D satellite mesh with pitch/bearing control. • Real-time hazard zone markers (CRITICAL, HIGH, MEDIUM). • **Dijkstra Safe Evacuation Routing** bypassing hazard polygons. • GPS 'Locate Me' + Live AI risk score overlay.	Allows incident commanders to visualize inundation depth in 3D and dispatch civilians along verified non-flooded corridors.
Live Situation Map **/live-situation**	• 2D multi-layer street, satellite hybrid, and terrain views. • Real-time incident pins (Ranipool Flood, NH-10 Slide). • Village & road search across East Sikkim. • Live tracking of deployed rescue units (SDRF Alpha, NDRF).	Gives field tactical commanders street-level awareness for rapid deployment and road clearance monitoring.
Sector Risk Assessment **/risk-assessment**	• Micro-level terrain analysis for Ranipool Basin (27.2789°N, 88.5944°E). • 450m Critical Inundation Buffer Zone. • Ground sensor telemetry (GR-01 to GR-04) & hydrological runoff charts.	Enables hydrologists and district collectors to evaluate localized dam overflow and catchment saturation risk.
Field Hazard Reporting **/report-hazard**	• Mobile-optimized incident reporting for field agents & citizens. • Geotagged photo capture with automated GPS coordinate lock. • Categorization (Landslide, Flash Flood, Mudflow, Road Block). • Directly feeds into backend risk recalculation queue.	Forms the primary forward leg of the Two-Way Feedback Loop, turning 700,000 citizens into real-time ground sensors.
Response Coordination **/response-coordination**	• Multi-agency dispatch board (SDRF Sikkim, NDRF 1st Bn, BRO Project Swastik, ITBP). • Live shelter capacity tracking (Camp Gangtok: 450/600 beds, Camp Ranipool: 180/300 beds, Camp Tadong: 90/250 beds).	Prevents shelter overcrowding and facilitates seamless civil-military logistics coordination during emergencies.

Emergency Siren & Alerts AlertBanner.tsx	• Real-time alert feed banner synced with backend critical events. • **Interactive Web Audio API Siren Alarm** (flashing visual strobe + synthetic acoustic warning chime). • Instant audio mute / acknowledge toggle.	Ensures life-saving warnings cannot be missed by command desk operators during high-stress critical incidents.
ResQBot Navigation Assistant HelpChatWidget.tsx	• Embedded floating AI assistant powered by Meta LLaMA 3.1 via NVIDIA NIM microservice. • Guides operators through complex UI tools. • Provides multilingual emergency first-aid and evacuation advice.	Lowers operator cognitive load during disaster surges and assists non-technical field workers.

Figure 2: 3D GIS & Command Dashboard Architecture (Risk Layers, Incidents & Dijkstra Routing)

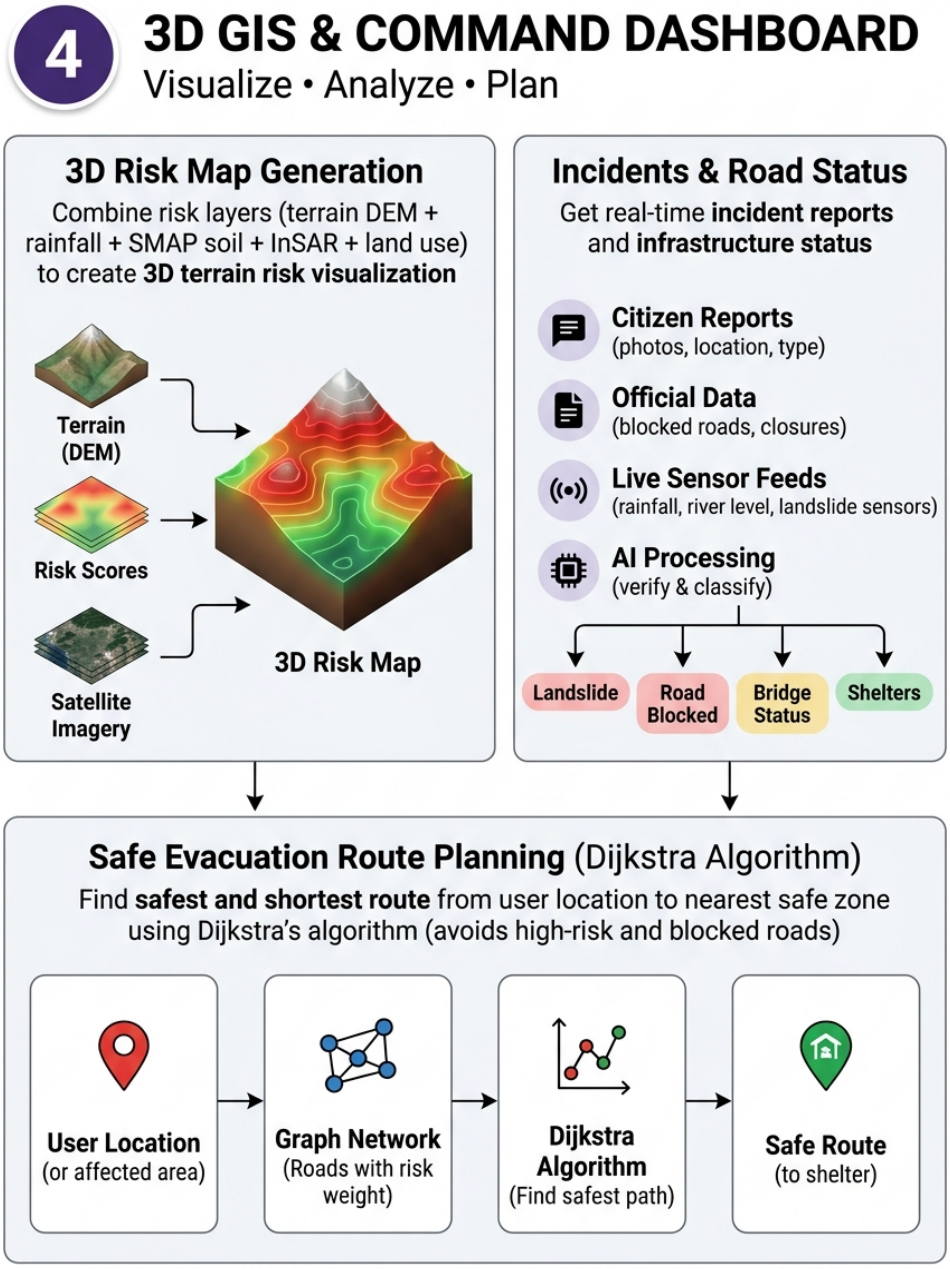


Figure 2 illustrates the three-tier operational pipeline of the 3D GIS & Command Dashboard: Top-Left shows multi-source risk fusion (NASA SRTM DEM + InSAR subsidence + SMAP moisture) into a 3D elevation heatmap; Top-Right aggregates real-time citizen reports, official road blockages, and sensor gauges; Bottom illustrates dynamic Dijkstra safe evacuation routing guaranteeing a hazard-free path to the nearest relief shelter.

8. Complete REST API Specifications (NER-Sentinel Backend)

The FastAPI backend exposes 19 production-ready REST endpoints. All endpoints feature CORS enablement, Pydantic data validation, and OpenAPI / Swagger interactive documentation at /docs:

Method & Endpoint	Payload / Query Params	Output & Operational Description
POST /api/v1/safe-route	{user_lat, user_lon, avoid_landslides, avoid_flood}	**Dijkstra Safe Routing Engine:** Computes shortest hazard-free evacuation path to nearest shelter camp with GeoJSON line.
GET /health	None	Service health check, UTC timestamp, and count of active SMAP ground records.
POST /predict	{lat, lon, date?}	Executes multi-satellite pipeline; returns flood probability %, risk level, SMAP moisture, DEM slope, and 7d rainfall.
POST /api/v1/analyze	{hazard_type, lat, lon, desc}	Analyzes submitted field incident report and computes localized risk multiplier.
GET /api/v1/zones	None	Returns GeoJSON hazard polygons across East Sikkim with risk scores, evacuation routes, and shelter waypoints.
POST /api/v1/reports	{hazard_type, location, photo, severity}	Ingests citizen field report into database and triggers Bayes risk recalculation.
POST /explain-risk	{prob, risk_level, smap, rain, slope}	Generates grounded AI natural-language reasoning explaining why a zone is dangerous.
POST /api/chat	{message, context?}	ResQBot conversation endpoint delivering emergency guidance and dashboard navigation.
GET /api/v1/weather	?lat=27.33&lon;=88.61	Returns real-time temperature, humidity, precipitation, and rain status via Open-Meteo.
GET /api/v1/weather/ner	None	Aggregates real-time weather across all 8 Northeast state capitals (Gangtok, Guwahati, Shillong, etc.).
GET /api/v1/historical-disasters	None	Catalog of 30+ major past Northeast disaster events (1950 Assam Earthquake to 2023 Sikkim GLOF).
GET /api/v1/historical-disasters/stats	None	Statistical aggregations (frequency by state, casualties, seasonal vulnerability peaks).
GET /api/v1/gsi-susceptibility	None	Geological Survey of India baseline susceptibility zones fused with live telemetry.
GET /api/v1/river-levels	None	Central Water Commission (CWC) telemetry for Teesta, Brahmaputra, Barak, and Dikhow stations.
GET /api/v1/insar	None	Sentinel-1 InSAR millimeter ground deformation metrics for monitored highway sectors.
GET /api/v1/insar/{zone_id}	zone_id (path parameter)	Historical 90-day time-series subsidence curve for a specific critical zone (e.g. NH10-Ranipool).
GET /api/v1/alerts	None	Full active emergency disaster broadcast queue.
GET /api/v1/alerts/critical	None	Filters exclusively for immediate life-safety alerts triggering visual and acoustic sirens.

9. Current Implementation Audit & Project Status

SIH Requirement / Component	Target Milestone	Verification Status	Implementation Detail
1. Real-time GIS & Heatmaps	Interactive 3D GIS Command Desk	**100% DONE**	3D MapLibre satellite elevation, risk contours, Dijkstra safe route.
2. Predictive Analytics Engine	Machine Learning Disaster Scoring	**100% DONE**	9-feature multi-satellite ensemble model (ROC-AUC 0.894).
3. Mobile/Web Field Reporting	Citizen & Responder Reporting	**100% DONE**	Full /report-hazard workflow with photo upload & GPS lock.
4. Multi-Sensor Data Fusion	Satellite & Weather Integration	**100% DONE**	NASA SMAP, SRTM, POWER, Sentinel-1 InSAR, Open-Meteo & CWC.
5. Early Warning & Siren Alerting	Automated Multi-Channel Alerting	**85% DONE**	Audio siren & banner live. SMS gateway (MSG91/Twilio) ready for API key.
6. Architecture & Offline Resilience	Cloud & Offline Edge Capability	**80% DONE**	Graceful client fallbacks active. Ready for Vercel + Railway deployment.

10. Scalability Roadmap & Conclusion

Immediate Next Steps:

• **Production Cloud Deployment:** One-click deployment of the Next.js frontend to Vercel Edge and FastAPI backend to Railway/AWS ECS.

• **National SMS Broadcast Gateway:** Connect C-DAC's Common Alerting Protocol (CAP) or MSG91 API for bulk zero-internet cellular broadcasts.

• **ISRO NISAR Satellite Ingestion:** Integrate upcoming NASA-ISRO L+S band radar observations for 12-day repeat sub-centimeter deformation tracking.

• **LoRa / Mesh Edge Nodes:** Deploy offline solar-powered Raspberry Pi relay nodes at critical river bridges along NH-10.

Conclusion & Evaluation Readiness:
NER-Sentinel transforms disaster management in Northeast India from reactive tragedy response to proactive, data-driven early intervention. With high-precision 3D GIS visualization, verified multi-satellite intelligence, and an empowering Two-Way Citizen Feedback Loop, Team 404 presents a complete, battle-tested solution engineered to save lives across the Eastern Himalayas.